

Sanjay Sakthirajan

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Summary

M.S. candidate in Data Science for Operations Research at the University of Minnesota with experience in data analytics, machine learning, and AI-powered applications. Skilled in Python, SQL, statistical analysis, and building end-to-end data solutions using machine learning and LLMs. Experienced in analyzing operational data and process performance metrics to identify bottlenecks and support optimization initiatives, contributing to an increase in production throughput from 20 to 22 units per shift. Built AI- and data-driven projects including an LLM-powered invoice extraction system, predictive machine learning models, and recommendation systems to generate actionable insights and support data-driven decision-making.

Employment history

Ganga Associates | Operations Intern

Jun 2022 - Sep 2022

- Analyzed production and operational data through time studies and workflow analysis, contributing to process improvements that increased line output from 20 to 22 units per shift.
- Performed workflow and capacity analysis with Excel pivot tables and Python data-cleaning scripts to pinpoint production bottlenecks, leading to recommended process changes that improved overall throughput
- Collected and analyzed production performance metrics using Python (pandas) and Excel dashboards, providing insights that guided operational decisions and streamlined scheduling processes
- Collaborated with production and engineering teams using shared Power BI reports and Python-generated data summaries to investigate process issues, which helped implement optimizations that smoothed the production flow

PSG Industries | Industrial Trainee

Aug 2020 - Jun 2024

- Collected, organized, and analyzed manufacturing performance data with Excel and time-study methods, using Python and Pandas for deeper analysis, which revealed several process-efficiency opportunities
- Evaluated workflow and process performance metrics using statistical analysis in Python, uncovering key inefficiencies that guided continuous-improvement projects
- Developed process maps and operational documentation to improve visibility into manufacturing workflows.
- Collaborated with production teams to monitor performance trends via Excel dashboards and Python scripts, identifying optimization opportunities that were added to the improvement backlog

Skills

SQL Databases, Machine Learning, Python, R, Matlab, pandas, Numpy, Matplotlib, Scikit-Learn, Data Cleaning, Exploratory Data Analysis, Statistical Analysis, Regression Analysis, Power BI, Microsoft Excel, Git, GitHub, Jupyter Notebook, PyCharm, AWS

Education

Master's, Data Science for operations Research

University of Minnesota-Twin Cities (GPA: 3.87/4.00)

Sep 2025 - Dec 2026

Minneapolis, Minnesota

- Coursework:** Data Science for Operations Research, Regression Analysis, Statistical Analysis, Optimization, Quality and Reliability, Project Management, Machine Learning, Analytics for Data Driven Decision Making, Decision Analysis, Game Theory

Bachelor's, Mechanical Engineering

PSG College of Technology (GPA: 3.17/4.00)

Jul 2020 - Jul 2025

Coimbatore

- Coursework:** Operations Research, Probability and Statistics, Manufacturing Systems, Quality Engineering, Supply Chain Management, Production Planning and Control, Engineering Economics

Certifications

- AWS Cloud Practitioner: AWS Certified Cloud Practitioner (CLF-C02) | Amazon Web Services | Jul 2026

Projects

Invoice Data Extractor

- Built an end-to-end AI-powered invoice extraction system using Claude API, Python, Pydantic, and MySQL to convert unstructured PDF invoices into structured data.
- Designed schema-based information extraction with Pydantic models to accurately extract invoice details, vendor information, and line items from diverse invoice formats.
- Developed a relational MySQL database to store extracted invoices and associated line items, enabling efficient querying and structured data management.
- Built and deployed an interactive Streamlit application for invoice upload, automated extraction, database integration, and real-time results visualization.
- Evaluated extraction quality using field-level accuracy and consistency validation to assess the reliability of LLM-generated outputs.

NBA Championship Predictor

- Designed an end-to-end predictive analytics pipeline by collecting 25 seasons of NBA team statistics using the NBA API and storing the data in a MySQL database.
- Engineered advanced basketball performance metrics and trained machine learning models using scikit-learn to predict NBA championship probabilities.
- Performed feature engineering and model evaluation to identify the statistical factors most strongly associated with championship success.
- Developed and deployed an interactive Streamlit application that enables users to explore championship predictions and team performance metrics in real time.

Netflix Recommendation App

- Developed a content-based movie recommendation system using TF-IDF, cosine similarity, and scikit-learn to generate personalized movie recommendations.
- Performed data preprocessing, feature engineering, and text vectorization on movie metadata to improve recommendation quality and similarity matching.
- Utilized Python and pandas to clean, transform, and analyze movie datasets for content-based recommendation modeling.
- Developed and deployed an interactive Streamlit application that delivers real-time movie recommendations based on user-selected titles.